

Cryptocurrency Option Price Discovery

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Abstract

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1 Introduction

Theoretical studies, notably those by Black (1975) and Easley et al. (1998), suggest that options markets possess informational advantages over their underlying markets, with price discovery occurring predominantly in options markets rather than spot markets. However, empirical research has yet to reach a consensus on this matter.

Certain studies provide supporting evidence: Ryu and Yang (2018) demonstrates that options trading can precipitate changes in stock prices, whilst Chakravarty et al. (2004) indicates that novel information regarding stock prices frequently manifests initially in options prices.

Conversely, research by Chan et al. (2002) and Muravyev et al. (2013) contends that options prices neither yield additional information beyond stock prices nor contribute substantially to price discovery. Furthermore, extensive empirical investigations, including works by Fleming et al. (1996), Chakravarty et al. (2004), Hu (2014), Chan et al. (2002), and Cheng and Zhou (2013), suggest that various characteristics of options contracts and markets – including trading costs, leverage, trading volume, bid-ask spreads, investor sentiment, macroeconomic announcements, and trading restrictions – influence the options market’s contribution to price discovery.

Given the significance of options price-implied information in empirical research and industry practices, particularly in forecasting, identifying the principal factors affecting price discovery in options markets could enhance forecasting accuracy and investment decisions by facilitating the selection of contracts that more efficiently incorporate information.

In this study, we employ the Information Leadership Share (ILS) methodology developed by Putniņš (2013), applying it to one-minute option price data for options written on Bitcoin (BTC) and Ethereum (ETH) cryptocurrencies to quantify the options markets’ role in price discovery.

Within cryptocurrency markets, whilst Ruan and Streltsov (2022) has established that perpetual futures markets influence price discovery in spot markets, a crucial question remains unaddressed: do options markets contribute to price discovery in cryptocurrency spot markets?

2 Literature Review

2.1 Motivation

3 Methodology

This is a detailed introduction on steps to Investigate Whether Option Prices Impact Price Discovery of the Underlying Price on cryptocurrency market.

3.1 Brief Steps

To determine the contribution of option prices to the price discovery of the underlying asset, the following steps can be followed.

Calculate the Option Implied Underlying Price Use an option pricing model (e.g., Black Scholes Merton (BSM)) to derive the implied cryptocurrency price I_t from the observed option price C_t :

$$I_t = f_V^1(C_t; \sigma),$$

where σ is the volatility parameter. Estimate σ using lagged cryptocurrency and option prices (e.g., 30 minute lag) to avoid circularity:

$$\hat{\sigma}_{tk} = f_{\sigma}^1(S_{tk}; C_{tk}).$$

Ensure that I_t and S_t (observed cryptocurrency price) are driven by a common stochastic factor (the efficient price V_t).

Model the Lead Lag Relationship Between Prices Using VECM Apply a Vector Error Correction Model (VECM) to the implied price I_t and the observed cryptocurrency price S_t :

$$\Delta p_{1,t} = \alpha_1(p_{1,t1} + \beta p_{2,t1}) + \text{lagged terms} + \epsilon_{1,t},$$

$$\Delta p_{2,t} = \alpha_2(p_{1,t1} + \beta p_{2,t1}) + \text{lagged terms} + \epsilon_{2,t}.$$

$p_{1,t}$: Log of implied price, $p_{2,t}$: Log of observed price. α_1, α_2 : Error correction coefficients indicating the speed of adjustment to equilibrium. The error correction term $(p_{1,t1} + \beta p_{2,t1})$ captures the long

term equilibrium relationship.

Estimate the Variance Covariance Matrix Compute the residuals $\epsilon_{1,t}$ and $\epsilon_{2,t}$ from the VECM and their variance covariance matrix Ω :

$$\Omega = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Perform Cholesky decomposition of Ω to orthogonalize the residuals:

$$M = \begin{bmatrix} m_{11} & 0 \\ m_{12} & m_{22} \end{bmatrix}.$$

Calculate Information Share (IS) Use the orthogonalized residuals and the error correction coefficients to compute the Information Share (IS) :

$$IS_1 = \frac{(\gamma_1 m_{11} + \gamma_2 m_{12})^2}{(\gamma_1 m_{11} + \gamma_2 m_{12})^2 + (\gamma_2 m_{22})^2},$$

$$IS_2 = \frac{(\gamma_2 m_{22})^2}{(\gamma_1 m_{11} + \gamma_2 m_{12})^2 + (\gamma_2 m_{22})^2}.$$

$\gamma_1 = \frac{\alpha_2}{\alpha_2\alpha_1}$, $\gamma_2 = \frac{\alpha_1}{\alpha_1\alpha_2}$. IS_1 : Contribution of implied prices to price discovery. IS_2 : Contribution of observed prices to price discovery.

Calculate Component Share (CS) Use the error correction coefficients to compute the Component Share (CS) :

$$CS_1 = \frac{\alpha_2}{\alpha_2\alpha_1}, \quad CS_2 = \frac{\alpha_1}{\alpha_1\alpha_2}.$$

CS_1 : Reflects the noise level in the implied price relative to the observed price.

Information Leadership We can then calculate the information leadership as IL_1 and IL_2 as

$$IL_1 = \left| \frac{IS_1}{IS_2} \cdot \frac{CS_2}{CS_1} \right|$$

$$IL_2 = \left| \frac{IS_2}{IS_1} \cdot \frac{CS_1}{CS_2} \right|$$

and the information leadership shares as ILS_1 and ILS_2 as

$$ILS_1 = \frac{IL_1}{IL_1 + IL_2}$$

$$ILS_2 = \frac{IL_2}{IL_1 + IL_2}$$

Interpret the Results Compare IS_1 and IS_2 : A higher IS_1 indicates that the option implied prices contribute more to price discovery. Use Component Share (CS) to assess the noise in each price series. Option prices are considered to have a significant price discovery impact if $IS_1 > IS_2$ and noise levels are low.

The details of each step are discussed in the following sections from Section 3.2 to 3.4.

3.2 Option Implied Underlying Price

The option implied underlying price is derived to represent the efficient cryptocurrency price V , which serves as the theoretical state variable underlying the observed cryptocurrency price S_t and the observed call option price C_t . This calculation is crucial for using option prices to infer the underlying cryptocurrency price and analyse how options contribute to price discovery.

The process is based on option pricing models (e.g., the Black-Scholes-Merton (BSM) model). The steps can be outlined as follows:

- **Observed cryptocurrency Price and Pricing Error:** The observed cryptocurrency price S_t at time t is modeled as:

$$S_t = V_t + e_{s,t},$$

where V_t is the efficient cryptocurrency price (state variable); $e_{s,t}$ is a zero-mean covariance-stationary error term representing microstructural frictions (e.g., bid-ask spreads, noise trading).

- **Option Pricing Model:** The observed call option price C_t is related to V_t through a theoretical option pricing model f , such as the BSM model:

$$C_t = f(V_t; \sigma),$$

where the σ is the volatility of the underlying asset return.

- **Inversion to Obtain Implied cryptocurrency Price:** The implied cryptocurrency price I_t is calculated by inverting the option pricing model with respect to V_t :

$$I_t = f_V^{-1}(C_t; \sigma),$$

where f_V^{-1} is the inverse function of the option pricing model.

- **Estimating Volatility σ :** Since σ is not directly observable, it must be estimated. The volatility is estimated as a function of cryptocurrency and option prices:

$$\hat{\sigma}_t = f_\sigma^{-1}(S_t; C_t).$$

Directly using S_t to estimate σ would tautologically make I_t equal to S_t . To avoid this, the paper uses the past k minute realized volatility based on the 1-min underlying high frequency return. At every minute t , we calculate the standard deviation of the return from $t - k$ to t as the $\hat{\sigma}_{t-k}$. This ensures that the implied cryptocurrency price is based on very recent historical information and avoids circularity.

- **Choice of Lag Length:** The lag k must be long enough for the errors $e_{s,t}$ and $e_{s,t-k}$ to be uncorrelated, ensuring that the estimation process is not biased by contemporaneous noise. However, if k is too long, the assumption of constant volatility σ over time becomes unrealistic. Following Stephan and Whaley (1990), the paper uses a 30-minute lag, based on evidence that implied volatility can change intraday.
- **Cointegration Between Implied and Actual cryptocurrency Prices:** Since options and cryptocurrency markets trade around the same underlying asset, the implied cryptocurrency price I_t and the observed cryptocurrency price S_t are assumed to be driven by a common stochastic factor (the efficient price V_t):

$$V_t = V_{t-1} + u_t, \quad u_t \sim i.i.d.N(0, \sigma_u),$$

where V_t follows a random walk, and u_t represents innovations in the fundamental value.

The Relevant Literatures include the following.

Chakravarty et al. (2004) suggested using option pricing models to calculate the implied cryptocurrency price to avoid the limitations of put-call parity, which requires using call and put options with the same strike price and expiration date, thereby reducing the cross-sectional information. They highlighted that option pricing models allow incorporating features such as leverage, bid-ask spreads, and trade volume into the analysis.

Stephan and Whaley (1990) proposed using lagged cryptocurrency and option prices to estimate implied volatility, as contemporaneous estimates can suffer from noise and circularity.

Booth et al. (1999); Kim et al. (2009) proposed for the use of the Black-Scholes-Merton (BSM) model to derive implied cryptocurrency prices for European-style options, especially when the binomial tree model converges to the BSM model.

Stoll (1969) claimed that while the put-call parity formula is simpler and does not rely on a specific model, it limits the cross-sectional information by requiring matching call and put options. This paper opts for the BSM model to include all types of option contracts.

3.3 Price Discovery Measure

The Vector Error Correction Model (VECM) is a framework used to model the dynamic relationship between multiple time series that are cointegrated. In the context of price discovery, the VECM estimates the parameters that describe the interactions between two price series (e.g., the implied cryptocurrency price from options and the observed cryptocurrency price). These parameters are then used to calculate Information Share (IS) and Component Share (CS), which measure the relative contributions of each price series to innovations in the fundamental value.

But we need to investigate the price discovery by a sliding window method. The window can be selected as 20 days. Thus, in each window, we have at least $60 \times 24 \times 20 = 28800$ data samples. Hence, at each minute t , we calculate a price discovery measure using the past 20 days, and then rolling forward to next minute $t + 1$. We can then calculate a series of price discovery measures.

3.3.1 Vector Error Correction Model (VECM)

The VECM equations for two price series, $p_{1,t}$ and $p_{2,t}$, are as follows:

$$\Delta p_{1,t} = \alpha_1(p_{1,t1} + \beta p_{2,t1}) + \sum_{i=1}^n \gamma_i \Delta p_{1,ti} + \sum_{j=1}^n \delta_j \Delta p_{2,tj} + \epsilon_{1,t},$$

$$\Delta p_{2,t} = \alpha_2(p_{1,t1} + \beta p_{2,t1}) + \sum_{k=1}^q \phi_k \Delta p_{1,tk} + \sum_{m=1}^q \rho_m \Delta p_{2,tm} + \epsilon_{2,t}.$$

Where:

- $p_{1,t}$: Log price of the first asset (e.g., implied cryptocurrency price),
- $p_{2,t}$: Log price of the second asset (e.g., observed cryptocurrency price),
- Δ : First difference operator,
- α_1, α_2 : Error correction coefficients, indicating the speed of adjustment,
- β : Cointegration coefficient, capturing the longterm equilibrium relationship,
- $\epsilon_{1,t}, \epsilon_{2,t}$: Residuals (innovations) from the VECM,
- $\gamma_i, \delta_j, \phi_k, \rho_m$: Coefficients for lagged terms representing shortterm dynamics.

The term $p_{1,t1} + \beta p_{2,t1}$ represents the error correction term, which captures the longterm relationship between the two price series. The coefficients α_1 and α_2 indicate how each price series adjusts to deviations from the equilibrium.

3.3.2 Variance Covariance Matrix of Residuals

The residuals $\epsilon_{1,t}$ and $\epsilon_{2,t}$ represent the shortterm innovations in each price series. The variancecovariance matrix Ω of these residuals is defined as:

$$\Omega = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix},$$

where: σ_1^2 : Variance of $\epsilon_{1,t}$, σ_2^2 : Variance of $\epsilon_{2,t}$, ρ : Correlation between $\epsilon_{1,t}$ and $\epsilon_{2,t}$.

The Cholesky decomposition of Ω is used to orthogonalize the residuals. The decomposition expresses Ω as:

$$\Omega = MM',$$

where M is a lower triangular matrix:

$$M = \begin{bmatrix} m_{11} & 0 \\ m_{12} & m_{22} \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 \\ \rho\sigma_2 & \sigma_2\sqrt{1-\rho^2} \end{bmatrix}.$$

3.3.3 Information Share (IS)

The Information Share (IS) measures the relative contribution of each price series to the discovery of innovations in the fundamental value.

Formula for IS The IS for each price series is calculated as:

$$IS_1 = \frac{(\gamma_1 m_{11} + \gamma_2 m_{12})^2}{(\gamma_1 m_{11} + \gamma_2 m_{12})^2 + (\gamma_2 m_{22})^2},$$
$$IS_2 = \frac{(\gamma_2 m_{22})^2}{(\gamma_1 m_{11} + \gamma_2 m_{12})^2 + (\gamma_2 m_{22})^2}.$$

where γ_1 and γ_2 are weights derived from the VECM error correction coefficients α_1 and α_2 .

How γ_1 and γ_2 are Calculated The weights γ_1 and γ_2 are normalized versions of the error correction coefficients α_1 and α_2 . They are calculated as:

$$\gamma_1 = \frac{\alpha_2}{\alpha_2 \alpha_1},$$
$$\gamma_2 = \frac{\alpha_1}{\alpha_1 \alpha_2}.$$

These weights reflect the relative speed at which each price series adjusts to the longterm equilibrium. Intuitively, a larger α (in absolute value) indicates that the corresponding price series plays a stronger role in restoring the equilibrium, and thus contributes less to price discovery.

IS_1 represents the proportion of the fundamental value's innovations that are primarily reflected in price series 1 (e.g., implied cryptocurrency price). IS_2 represents the same proportion for price series 2 (e.g., observed cryptocurrency price). By construction, $IS_1 + IS_2 = 1$.

3.3.4 Component Share (CS)

The Component Share (CS) focuses on the noise levels in each price series and their relative contributions to price discovery.

Formula for CS The CS for each price series is calculated as:

$$CS_1 = \frac{\alpha_2}{\alpha_2 \alpha_1},$$

$$CS_2 = \frac{\alpha_1}{\alpha_1 \alpha_2}.$$

These formulas are identical to the weights γ_1 and γ_2 . Hence, the CS represents the level of noise in each price series relative to the other.

CS_1 measures the proportion of noise in price series 1 relative to price series 2. CS_2 measures the same for price series 2.

Unlike IS, CS does not directly measure the contribution to reflecting innovations in the fundamental value. Instead, it captures the relative noise levels in the two series.

3.4 Information Leadership Share (ILS)

To address the limitations of IS and CS when noise levels differ, Putniņš (2013) proposed the ILS, which combines IS and CS to measure the speed and degree of information integration:

$$IL_1 = \frac{IS_1}{IS_2} \cdot \frac{CS_2}{CS_1}, \quad IL_2 = \frac{IS_2}{IS_1} \cdot \frac{CS_1}{CS_2}.$$

The ILS is then normalized to sum to 1:

$$ILS_1 = \frac{IL_1}{IL_1 + IL_2}, \quad ILS_2 = \frac{IL_2}{IL_1 + IL_2}.$$

Values of $ILS_1 > 0.5$ indicate that the first price series leads in price discovery.

Information Leadership Following Putniņš (2013), we can then calculate the information leadership as IL_1 and IL_2 as

$$IL_1 = \left| \frac{IS_1}{IS_2} \cdot \frac{CS_2}{CS_1} \right|$$

$$IL_2 = \left| \frac{IS_2}{IS_1} \cdot \frac{CS_1}{CS_2} \right|$$

and the information leadership shares as ILS_1 and ILS_2 as

$$ILS_1 = \frac{IL_1}{IL_1 + IL_2}$$

$$ILS_2 = \frac{IL_2}{IL_1 + IL_2}$$

3.5 Key Insights from VECM in IS and CS Calculation

Error Correction Coefficients Determine Leadership The error correction coefficients α_1 and α_2 reflect the speed of adjustment of each price series to deviations from the equilibrium. A slower adjustment (smaller α) suggests a greater role in price discovery.

Variance Covariance Matrix Captures Innovations: The residuals from the VECM model ($\epsilon_{1,t}$ and $\epsilon_{2,t}$) and their covariance matrix Ω capture the innovations in the price series. The Cholesky decomposition of Ω is critical for orthogonalizing these innovations, which ensures that IS calculations are not biased by correlation between the residuals.

Combination of LongTerm and ShortTerm Dynamics: The error correction term in the VECM represents the longterm equilibrium relationship, while the lagged differences capture short-term dynamics. Both are crucial for determining the relative contributions of each price series to price discovery.

IS and CS Are Complementary: IS combines noise levels and leadership in incorporating new information, while CS focuses purely on noise levels. Together, they provide a more comprehensive view of price discovery.

In conclusion, VECM results are used to calculate IS and CS by extracting error correction coefficients (α_1, α_2) and the variancecovariance matrix (Ω) of residuals. IS measures the relative contribution of each price series to reflecting innovations in the fundamental value, using normalized coefficients γ_1 and γ_2 , derived from α_1 and α_2 . CS measures the relative noise levels in the two price series, directly using α_1 and α_2 .

The Relevant Literatures include the following.

Hasbrouck (1995) introduced the Information Share (IS) as a measure of price discovery. Highlighted the role of noise and information in determining IS.

Booth et al. (1999); Harris et al. (2002) proposed the Component Share (CS), which measures price series' noise levels and contributions to price discovery.

Yan and Zivot (2010) demonstrated that IS captures both noise and leadership in price discovery, while CS focuses on noise. They proposed combining IS and CS to account for both aspects.

Putniņš (2013) introduced the Information Leadership Share (ILS) to eliminate the noise effects in IS and CS, providing a cleaner measure of price discovery.

Patel et al. (2020) confirmed the robustness of ILS as a superior measure of price discovery using Monte Carlo simulations.

3.6 Key Implementation Issue

We need to collect the cryptocurrency option and underlying data for BTC and ETH. We shall first calculate the option implied underlying price. In this calculation, we select the following settings:

- We select **At-the-Money Call** options with $\frac{K}{S} \in [0.99, 1.01]$ and **30 day maturity** to calculate the ATM Call Discovery Measure
- We select **Out-of-the-Money Call** option with $\frac{K}{S} \in (1.01, 1.2]$ with **30 day maturity** to calculate the OTM Call Discovery Measure
- We select **In-the-Money Call** option with $\frac{K}{S} \in [0.8, 1)$ with **30 day maturity** to calculate the ITM Call Discovery Measure
- Furthermore, We select **At-the-Money Put** options with $\frac{K}{S} \in [0.99, 1.01]$ and **30 day maturity** to calculate the ATM Put Discovery Measure
- We select **Out-of-the-Money Put** option with $\frac{S}{K} \in (1, 1.2]$ with **30 day maturity** to calculate the OTM Put Discovery Measure
- We select **In-the-Money Put** option with $\frac{S}{K} \in [0.8, 1)$ with **30 day maturity** to calculate the ITM Put Discovery Measure

For ATM, OTM and ITM Discovery Measure, we calculate the average value of option implied underlying prices.

We need to identify the strongest Discovery Measure among the above six. We refer this as the Price Discovery Measure of the option.

The example Python code to apply VECM (Vector Error Correction Model), and calculate the IS, CS, IT, and ILS is as the file “InformationLeadershipShare.ipynb”.

3.7 What characteristics Contributes to the Price Discovery Measure

After identifying the Price Discovery Measure, we need to identify what option characteristics Contributes to it. To do this, we apply Random Forest, XGBoost, and NN to predict the Price Discovery Measure using the option statistical characteristics.

When considering all options (ATM, ITM, and OTM), we can calculate a variety of statistical characteristics. These characteristics, derived from option prices and implied volatilities, include expected moments, skewness, kurtosis, volatility, and more. Below is a summary of the key characteristics and their respective formulas.

- **Implied Volatility Surface.** Definition: The implied volatility across all strikes (ITM, ATM, and OTM options) and expirations, which forms the basis for calculating other statistical characteristics. Purpose: Describes how volatility varies with strike price and maturity. Formula: Numerically solve for implied volatility (σ) by equating market option prices to theoretical prices:

$$C_{\text{market}} = C_{\text{BS}}(S, K, T, r, \sigma_{\text{implied}})$$

Where: C_{market} : Market price of the option. C_{BS} : BlackScholes formula for call price. S, K, T, r : Spot price, strike price, time to expiration, and riskfree rate. σ_{implied} : Implied volatility (solved numerically).

- **RiskNeutral Probability Density Function (PDF)** Definition: The implied riskneutral probability density function (PDF) for the underlying asset's future price, derived from option prices. Purpose: Provides the marketimplied probabilities of different future price outcomes. Formula (BreedLitzenberger):

$$f(K) = e^{rT} \frac{\partial^2 C(K)}{\partial K^2}$$

Where: $f(K)$: Riskneutral PDF at strike K . $C(K)$: Call option price as a function of strike K . r : Riskfree rate. T : Time to maturity.

Practical Approximation (finite differences for the second derivative):

$$\frac{\partial^2 C(K)}{\partial K^2} \approx \frac{C(K + \Delta K) - 2C(K) + C(K - \Delta K)}{\Delta K^2}.$$

- **Expected Return Moments** These moments (M_n) describe the shape of the risk neutral distribution and are directly linked to option prices.

Second Moment (Variance) Definition The risk neutral variance of the underlying asset's returns. Formula:

$$M_2 = \frac{2e^{rT}}{T} \left[\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right] - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2 \quad (1)$$

Where:

- $F = Se^{(r-q)T}$ (forward price). T is the maturity time in years.
- K_0 is the ATM strike closest to F (or interpolated).
- The subtraction term adjusts for the drift in log-returns.

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Third Moment (Cubic Moment, Related to Skewness) Definition: The third central moment $M_3 = e^{rT} \mathbb{E}^Q[R^3]$, which captures the asymmetry of the risk-neutral return distribution. A positive M_3 indicates right-skewness (fatter right tail, e.g., upside potential in bull markets), while negative suggests left-skewness (downside risk, common in equities and crypto during crashes). It's the fair value of a “cubic contract” paying R^3 , spanned by a portfolio of options.

$$M_3 = e^{rT} \int_0^\infty \frac{6 \ln\left(\frac{K}{F}\right) - 3 \left[\ln\left(\frac{K}{F}\right)\right]^2}{K^2} O(K) dK \quad (2)$$

Where:

- $O(K) = P(K)$ if $K < F$ (OTM put price),
- $O(K) = C(K)$ if $K \geq F$ (OTM call price),
- $P(K)$: Put option price with strike K ,
- $C(K)$: Call option price with strike K ,
- The integrand weights $6 \ln(K/F) - 3[\ln(K/F)]^2$ derive from Taylor expansion of the cubic payoff around the forward.

Fourth Moment (Quartic Moment, Related to Kurtosis) Definition: The fourth central moment $M_4 = e^{rT} \mathbb{E}^Q[R^4]$, measuring the tail heaviness and peakedness of the risk-neutral return distribution relative to a normal (kurtosis=3). Values >3 indicate leptokurtosis

(fat tails, common in crypto due to jumps), implying higher probability of extreme events. It's the fair value of a "quartic contract" paying R^4 .

$$M_4 = e^{rT} \int_0^\infty \frac{12 \left[\ln \left(\frac{K}{F} \right) \right]^2 - 4 \left[\ln \left(\frac{K}{F} \right) \right]^3}{K^2} O(K) dK \quad (3)$$

Where $O(K)$, $P(K)$, and $C(K)$ are defined as above. The integrand weights $12[\ln(K/F)]^2 - 4[\ln(K/F)]^3$ come from the quartic payoff expansion.

Rationale and Derivation Insights:

- Analogous to the third moment, this spans the R^4 payoff with higher-powered logarithmic weights, emphasizing extreme strikes for tail measurement.
- The structure builds on lower moments: Variance uses $2/K^2$, skewness adds linear log terms, kurtosis quadratic and cubic logs.
- Positive contributions from both tails; in fat-tailed markets (e.g., crypto), deep OTM options drive M_4 upward, reflecting jump risks.
- Assumes the first moment $\mathbb{E}^Q[R] = 0$ (true for log-returns under risk-neutrality with no drift bias); otherwise, adjust for centering.
- Link to previous: These moments enable full distribution reconstruction (e.g., via Gram-Charlier expansion) or risk measures like Value-at-Risk.

Implied (Shannon) entropy under the risk-neutral measure is

$$\text{Entropy} = - \int_0^\infty f(K) \ln(f(K)) dK \quad (4)$$

Where $f(K)$ is the risk-neutral PDF of S_T , normalized such that $\int_0^\infty f(K) dK = 1$. This yields a positive (or less negative) value for higher uncertainty.

Fit a Parametric Distribution:

- **Lognormal Approximation** (easiest for quick calc): Assume $S_T \sim \text{Lognormal}(\mu, \sigma^2)$, where $\mu = \ln(F) - \sigma^2/2$ (risk-neutral mean), and $\sigma^2 = M_2T$ from the variance formula.
- PDF: $f(K) = \frac{1}{K\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln K - \mu)^2}{2\sigma^2}\right)$ for $K > 0$.
- **Probability of Expiry In the Money (ITM Probability)** Under the risk-neutral measure Q , it represents $\mathbb{E}^Q[\mathbb{I}_{S_T > K}]$ for calls (or $\mathbb{E}^Q[\mathbb{I}_{S_T < K}]$ for puts), where S_T is the underlying price at

expiration T , and K is the strike. This is the probability that the option has positive intrinsic value at expiry (ITM: $S_T > K$ for calls, $S_T < K$ for puts). It's equivalent to the option's delta in certain normalizations and is implied by market prices via model inversion. In crypto (e.g., BTC options), high volatility can make ITM probabilities sensitive to σ , signaling market expectations of large moves.

For call options:

$$P(\text{ITM}) = N(d_2) \quad (5)$$

For put options:

$$P(\text{ITM}) = 1 - N(d_2) = N(-d_2) \quad (6)$$

Where:

$$d_2 = \frac{\ln(S/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} \quad (7)$$

And $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz$ (standard normal CDF), where spot price S , strike K , time to expiration T , risk-free rate r , dividend yield q (set to 0 for crypto), implied volatility σ .

- Tail Risk Metrics RightTail and LeftTail Probabilities: Definition: Probabilities of extreme upward or downward price moves. Formula (approximated from PDF):

$$P(\text{RightTail}) = \int_{K_{\text{high}}}^{\infty} f(K) dK = 1 - F(K_{\text{high}}) \quad (8)$$

$$P(\text{LeftTail}) = \int_0^{K_{\text{low}}} f(K) dK = F(K_{\text{low}}) \quad (9)$$

Where:

- $f(K)$: Risk-neutral PDF of the underlying price S_T at expiration T ,
- $F(K)$: Corresponding CDF,
- $K_{\text{high}} > F$ (e.g., $F + 3\hat{\sigma}\sqrt{T}$, where $\hat{\sigma}$ is ATM implied volatility)
- $K_{\text{low}} < F$ (e.g., $F - 3\hat{\sigma}\sqrt{T}$),
- $F = Se^{(r-q)T}$ (forward price).

Simple OTM Option-Based Approximation: Use Black-Scholes CDF with implied σ from ATM options: $P(\text{RightTail}) = N(-d_2(K_{\text{high}}))$, where $d_2 = \frac{\ln(F/K_{\text{high}}) - (\sigma^2/2)T}{\sigma\sqrt{T}}$ (note sign flip for survival).

- **Volatility Skew Definition:** Measures the slope of implied volatility across strikes, indicating the market's expectations of asymmetric risk. Formula (Slope):

$$\text{Skew} = \frac{\sigma(K_{\text{OTM}}) - \sigma(K_{\text{ITM}})}{K_{\text{OTM}} - K_{\text{ITM}}}$$

Where $\sigma(K)$ is the implied volatility at strike K .

Aggregate OTM and ITM Implied Vols: Direct use of single $\sigma(K_{\text{OTM}})$ and $\sigma(K_{\text{ITM}})$ (e.g., selecting specific deltas like 25-delta OTM) is common but sensitive to choice. Aggregation mitigates noise from individual quotes.

Shall We Calculate the Average? Yes, averaging is a standard and recommended approach for robustness, especially with noisy data (e.g., crypto options). We can use simple average: $\bar{\sigma}_{\text{OTM}} = \frac{1}{N_{\text{OTM}}} \sum_{i \in \text{OTM}} \sigma(K_i)$. However, weighted averages provide better accuracy. But due to data limitation, we might not be able to calculate the weighted average.

- **Volume-Weighted:** Weight by trading volume or open interest to emphasize liquid strikes (reduces bid-ask noise).
 - **Vega-Weighted:** Weight by option vega ($\partial C / \partial \sigma$), as it reflects sensitivity to vol changes: $\bar{\sigma}_{\text{OTM}} = \frac{\sum_i \sigma(K_i) \cdot \text{vega}_i}{\sum_i \text{vega}_i}$.
 - **Moneyness-Binned:** Group into bins (e.g., 10% OTM, 20% OTM) and average within bins, then use bin representatives.
- **Volatility Smile Curvature Definition:** Measures the curvature of the implied volatility surface, indicating the presence of fat tails.

$$\text{Curvature} = \frac{\sigma(m_{\text{OTM}}) + \sigma(m_{\text{ITM}}) - 2\sigma(m_{\text{ATM}})}{(m_{\text{OTM}} - m_{\text{ATM}})^2} \quad (10)$$

Where:

- $m = \ln(K/F)$ (log-moneyness, with $F = S e^{(r-q)T}$ the forward price),
- $m_{\text{ATM}} \approx 0$ (e.g., $K_{\text{ATM}} \approx F$),
- $m_{\text{OTM}} > 0$ (e.g., delta=0.25 call), $m_{\text{ITM}} < 0$ (symmetric, e.g., delta=0.75 call or equivalent put via parity),

- For asymmetric cases, use a general second derivative approx: $\sigma''(m) \approx \frac{2[\sigma(m_+) - \sigma(0) - \sigma(m_-) + \sigma(0)]}{(m_+)^2 + (m_-)^2}$
or fit a quadratic regression.

We need to identify the appropriate ones from the above list to construct the training and testing dataset for the prediction.

After the prediction, the feature importance shall be calculated. This will be discussed later.

4 Empirical Analysis

Appendix

A xxxxxx

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